

A Method of Removing Ocular Artifacts from EEG using Discrete Wavelet Transform and Kalman Filtering

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Abstract—Electroencephalogram (EEG) is a noninvasive method to record electrical activity of brain and it has been used extensively in research of brain function due to its high time resolution. However raw EEG is a mixture of signals, which contains noises such as Ocular Artifact (OA) that is irrelevant to the cognitive function of brain. To remove OAs from EEG, many methods have been proposed, such as Independent Components Analysis (ICA), Discrete Wavelet Transform (DWT), Adaptive Noise Cancellation (ANC) and Wavelet Packet Transform (WPT). In this paper, we present a novel hybrid de-noising method which uses Discrete Wavelet Transform (DWT) and Kalman Filtering to remove OAs in EEG. Firstly, we used this method on simulated data. The Mean Squared Error (MSE) of DWT-Kalman method was 0.0017, significantly lower compared to results using WPT-ICA and DWT-ANC, which were 0.0468 and 0.0052, respectively. Meanwhile, the Mean Absolute Error (MAE) using DWT-Kalman achieved an average of 0.0052, which also performed better than WPT-ICA and DWT-ANC, which were 0.0218 and 0.0115, respectively. Then we applied the proposed approach to the raw data collected by our prototype three-channel EEG collector and 64-channel Braincap from BRAIN PRODUCTS. On both data, our method achieved satisfying results. This method does not rely on any particular electrode or the number of electrodes in certain system, so it is recommended for ubiquitous applications.

Keywords—discrete wavelet transform; EEG; ocular artifacts; Kalman filtering

I. INTRODUCTION

Electroencephalogram (EEG) is a method to record electrical potential generated by brain activities using electrodes attached to scalp with precise placement according to 10-20 national standard system. Since it is a noninvasive method, EEG has a wide range of applications in the following aspects: cognitive states [1-3] including attention, alertness, mental workload, etc.; brain computer interface (BCI) and neurological disorders such as Epilepsy, Attention Deficit Hyperactivity Disorder, Autism Spectrum Disorder [4] and so on. Unfortunately, EEG signals are easily contaminated by extrinsic noise from equipment and intrinsic physiological

artifacts produced by subjects such as Ocular Artifacts (OAs), Electromyography (EMG) artifacts and Electrocardiogram (ECG) artifacts. Among all these artifacts, OA is the most commonly encountered one, which has much larger magnitude compared to EEG. The distorted EEG can mislead interpretation clinically [5], so removing OAs from EEG becomes an essential work before analysis.

In early days, subjects were required to close their eyes in order to reduce the appearance of OAs. However, this method cannot eliminate OAs completely due to the fact that even when subjects close their eyes, eye movements can still be recorded.

At present, many conventional signal processing algorithms are used to solve the problem. Regression analysis methods have been used to correct EEG in both time domain [6, 7] and frequency domain [8, 9]. This type of methods is implemented based on the assumption that OAs and EEG are linearly uncorrelated. Although this algorithm is simple, practical Electrooculography (EOG) recordings certainly include EEG and vice versa, hence, this bidirectional cross-interference inevitably affects de-noising results.

Adaptive filter is also used to remove OAs from EEG by LMS algorithm [10] and VSSLMS algorithm [11]. These methods usually take the output of adaptive filter as OA interference, then subtract it from EEG [12]. Adaptive filter can automatically adjust parameters under the condition that priori knowledge of noise is unknown, and de-noise adaptively. The precondition of using adaptive filtering is an extra EOG reference channel.

Several documents [13-16] use independent components analysis (ICA) to attenuate OAs in EEG. ICA is a kind of blind signal separation method, it does not need EOG reference channel. However, the independent components separated by ICA are of uncertainty in terms of sorting, which leads to visual classification of components [17, 18]. In addition, how to determine the number of independent components effectively and reducing computation complexity are problems which need to be considered [19].

There are many people using wavelet transformation to eliminate OAs because of its multi-resolution property [20-23]. In general, EEG is decomposed by wavelet functions into several parts, then the decomposition coefficients are compared with some threshold and processed in order to reconstruct true EEG. The deficiency of this approach is that the decomposition level and threshold are hard to be determined.

In order to achieve better result, preferable implementation is to combine different methods. References [4, 24, 25] bond wavelet transform and ICA to remove OAs whereas wavelet transform and adaptive filter are applied in our previous work [26, 27]. There are many other methods used to remove OAs from EEG, such as these based on Hilbert-Huang Transform [28], singular value decomposition [29], support vector machine [30-32] and so on.

In this paper, we propose a novel hybrid approach of OAs removal, which combines DWT and Kalman filtering. To demonstrate the effectiveness of this method, we quantitatively evaluate the mean squared error (MSE) and the mean absolute error (MAE) of different models based on simulated data, then apply it to the data collected by both our prototype three-channel EEG collector and 64-channel braincap produced by BRAIN PRODUCTS, respectively. This paper is arranged as follows: research background of EEG and some methods of OAs removing are stated in the first part. In the second part, two models used for comparison are presented and our approach is described in detail. In the third part, some experiments and performance evaluation are made. Lastly, the paper ends up with discussion.

II. MODELS AND METHODOLOGY

Introduction has described research background and some OAs removal methods. In this section, we thoroughly explain our discrete wavelet transform and Kalman filtering (DWT-Kalman) approach, along with two other popular models for comparison: wavelet packet transform and independent components analysis (WPT-ICA), as well as discrete wavelet transform and adaptive noise cancellation (DWT-ANC).

EEG recordings are frequently polluted by ocular artifacts. There is a consensus among researchers that cerebral activities and eye activities are generated by different sources. Consequently, OAs within EEG recordings can be treated as additive noise [33]. So we make a general assumption (1), where $EEG_{rec}(t)$ is recorded EEG signal and $EEG_{true}(t)$ is true EEG signal without interference, $OA(t)$ represents the ocular artifacts.

$$EEG_{rec}(t) = EEG_{true}(t) + OA(t) \quad (1)$$

A. WPT-ICA Model

Basic idea underlying ICA is that we can get independent source components via analyzing observed signals in the case of source signals are uncharted. Multi-channel signal input is a prerequisite for application of ICA. However, the number of observed EEG recordings are finite in practice on account of

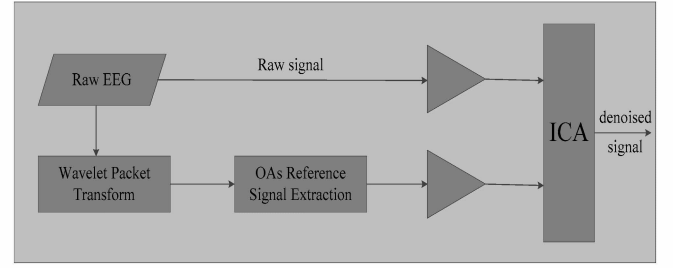


Fig. 1. Block diagram of WPT-ICA.

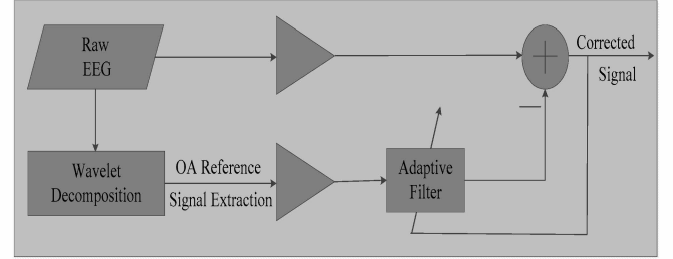


Fig. 2. Block diagram of DWT-ANC.

limited EEG channels. As an effective and precise frequency-based tool which can extract both time and frequency domain information in EEG signals, wavelet packet transform is well suited to extract reference OAs as the input of ICA. The block diagram of WPT-ICA model is shown in Fig. 1.

This model mainly contains two steps to recover true EEG. Initially, WPT is used to divide raw EEG into different wavelet packet coefficients in various resolution scales. Then, we select the soft threshold and apply it to the coarsest approximation coefficients for the reason that OAs are chiefly concentrated in low frequency band. The reference signal is then reconstructed using the approximation coefficients. Secondly, both reference signal and raw EEG are taken as input of ICA to eliminate OAs.

B. DWT-ANC Model

Adaptive filter is able to track the input signal time varyingly and acquire optimal output. It estimates interference effectively using reference signal, and then, the interference estimation is subtracted from raw EEG composed of desired signal and noise, finally the desired signal is obtained.

Since adaptive filter requires two inputs, reference noise is necessary, so the application of DWT offers possibility to employ adaptive filter in case of EEG recordings without EOG channel. The block diagram of DWT-ANC model is shown in Fig. 2.

The procedure of this model to eliminate OAs is as follows: Firstly, wavelet transform is employed to decompose raw EEG, then, we construct the reference input of adaptive filter after processing the wavelet decomposition coefficients by soft-thresholding. Finally, the output of adaptive filter based on recursive least-squares (RLS) algorithm is subtracted from raw EEG to acquire desired signal.

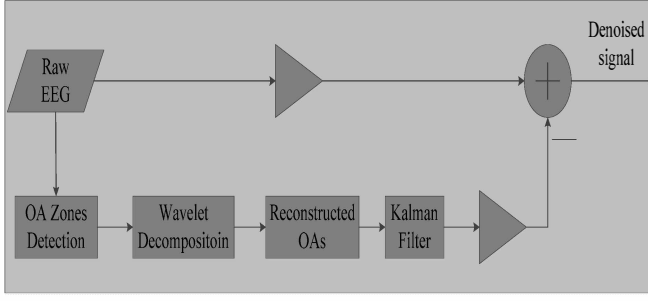


Fig. 3. Block diagram of DWT-Kalman filtering.

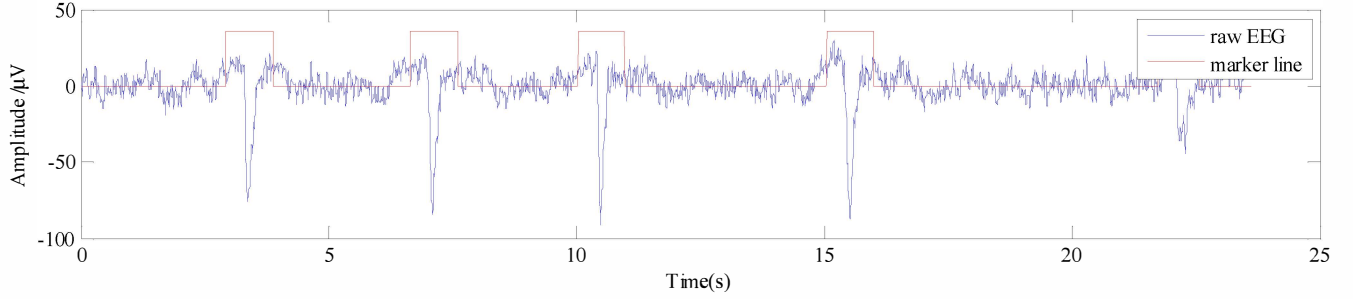


Fig. 4. Automatic detection of OA zones in EEG recordings.

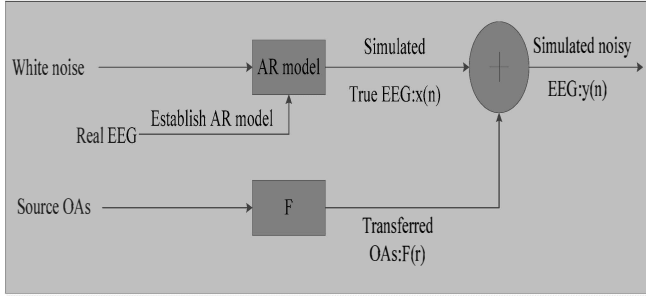


Fig. 5. General procedure for producing simulated EEG data.

C. DWT-Kalman Filtering Model

Two popular methods of correcting EEG have been described, then, we will explain our proposed model in detail for OAs removal. The model, in Fig. 3, contains four steps. At first, we detect the OA Zones in raw EEG; secondly, DWT is applied to the raw EEG in OA zones to reconstruct rough OAs approximations; thirdly, Kalman filtering is used to optimize the OAs approximations from the previous step; and lastly, the optimized OAs is subtracted from raw EEG.

As OAs just appear randomly in EEG recordings, we should not apply OAs removal methods to whole EEG epochs. Otherwise, much cerebral activities in non-OA zones will be affected. For the purpose of avoiding such problem, detection of OA regions is essential. Dealing with raw EEG only at OA areas can maximum preserve available EEG. S. Venkata [22] employs HAAR wavelet to decompose raw EEG, then obtains a step function with a falling edge for change in the state of eyes from open to closed and a step function with a rising edge for change in state of eyes from closed to open. However, based on the fact that OAs are several times stronger than EEG, we identify OA areas with a threshold, which is

determined by the mean value and standard deviation of EEG recordings without OAs. The mean value represents global average characteristic whereas the standard deviation describes degree of dispersion of the samples. The threshold is determined as in (2), where Th means the threshold, m and σ stand for mean value and standard deviation of the EEG signal without OAs, respectively. The factor k is tunable, which can be determined according to practical situation. In our experiments, we find that the optimal results will be obtained when $k = 4$.

$$Th = m \pm k \times \sigma \quad (2)$$

After Th is determined, we compare it with each EEG samples and segments beyond the threshold are marked. Then, the marked parts are extended to both forward and backward sides, thus OA regions are determined. Compared with the HAAR wavelet decomposition method, this approach sharply reduces computation complexity. Our research results suggest that this method identify OA regions accurately, as shown in Fig. 4.

Next step is extracting approximated OAs from OA zones. We choose db4 wavelet to divide OA zones into different resolution scales. Since OAs mainly concentrate in the low-frequency band, we should use coarse approximation coefficients to reconstruct approximated OAs. Because frequency overlap between EEG and interference events, such as the fact that some EEG signal in low-frequency band may be distributed in approximated OAs, it is unreasonable to discard approximated OAs directly. Now, cerebral signal in approximated OAs needs to be filtered out to obtain pure OAs. Then they will be subtracted from raw EEG.

We use AR model to denote the relationship between pure OAs and OAs mixed with EEG, as expressed in (3) and (4), where A_{k-1} is AR model parameters and $\omega(k-1)$ is a zero mean Gaussian noise with covariance Q_k ; $x(k)$ is pure OAs, $v(k)$ is EEG in low-frequency band and $z(k)$ is the approximated OAs. k is an integer and indicates discrete equidistant time point. In order to get right prediction error, we set the order of AR model to 4.

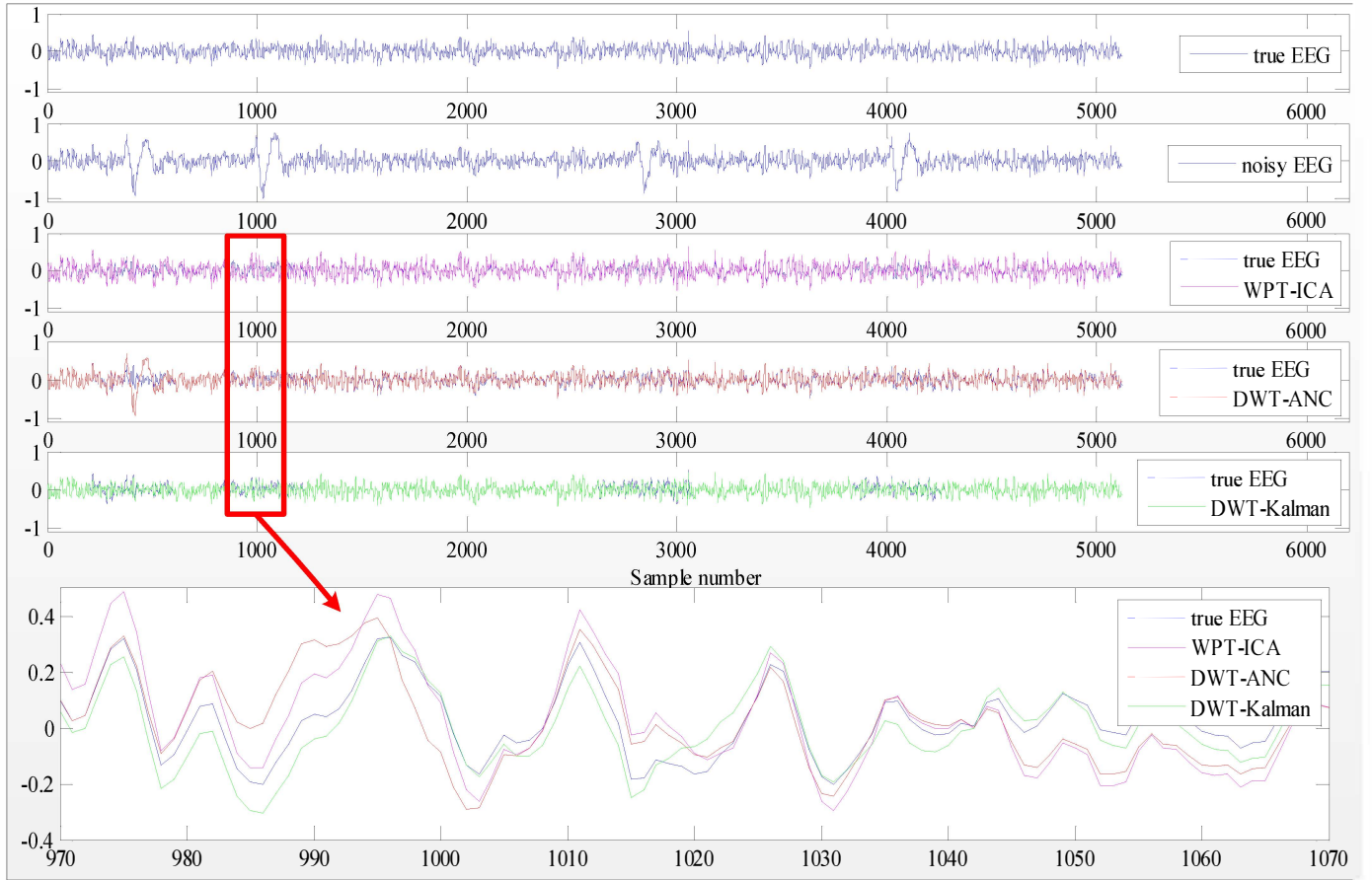


Fig. 6. First line: simulated true EEG. Second line: simulated noisy EEG. Third line: true EEG and corrected EEG by WPT-ICA. Fourth line: true EEG and corrected EEG by DWT-ANC. Fifth line: true EEG and corrected EEG by our proposed model. Bottom line: detail display of part of three lines above the bottom line.

$$\begin{aligned} x(k) &= a_1 x(k-1) + a_2 x(k-2) + \dots + a_p x(k-p) + \omega(k-2) \\ &= A_{k-1} X(k-1) + \omega(k-2) \end{aligned} \quad (3)$$

$$z(k) = x(k) + v(k) \quad (4)$$

From formulas (3) and (4), it is natural to choose Kalman filtering to estimate pure OAs because of its eminent tracking and estimation characteristics. Furthermore, Kalman filtering adopts recursive algorithm to produce optimal estimation, which is suitable for computer processing. The procedure of Kalman filtering is presented as (5), (6), (7), (8)

$$P'_k = A_k P_{k-1} A'_k + Q_{k-1}, \quad (5)$$

$$H_k = \frac{P'_k C_k^T}{C_k P'_k C_k^T + R_k}, \quad (6)$$

$$\hat{x}_k = A_k \hat{x}_{k-1} + H_k (z_k - C_k A_k \hat{x}_{k-1}), \quad (7)$$

$$P_k = (I - H_k C_k) P'_k. \quad (8)$$

The meaning of each variable is as follows. A_k is state variable gain matrix (AR model parameters), C_k presents observation matrix, Q_k means covariance of $\omega(k)$, R_k is covariance of $v(k)$, P_k indicates posteriori estimation error covariance, P'_k stands for priori estimation error covariance, H_k is Kalman gain, $\hat{x}_k$ is estimation of state variable at point k , and z_k means observed data (approximated OAs). Finally, we just subtract the output of Kalman filtering from raw EEG and clean EEG is obtained.

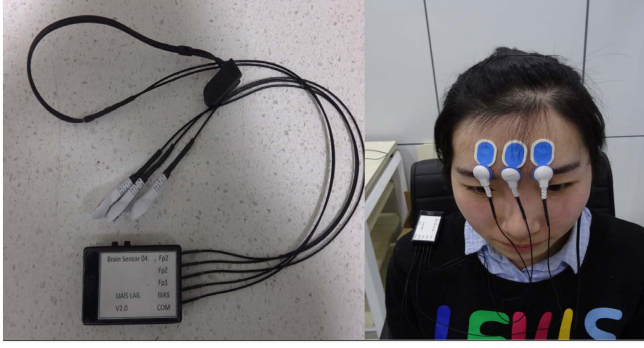


Fig. 7. Three-channel EEG collector.

III. EXPERIMENTS

In order to validate the effectiveness of our approach, experiments are necessary. We apply it to simulated data and real data to test its performance.

A. Simulated Data Test

The major drawback of using real EEG is that true EEG is unknown, so it is inconvenient to quantitatively evaluate the performance of this method. In order to solve this problem, we have used simulated data to perform a quantitative evaluation on accuracy of removing OAs from EEG. The simulated data is generated according to document [34] and general procedure is shown in Fig. 5. Fifty true EEG waveforms are simulated and normalized, each contains 5120 samples with sampling frequency 256Hz. Each EEG waveform includes four EOG pulses randomly selected from ten seed EOG pulses. The amplitude of each EOG is tuned by a uniformly distributed random factor between 3.2 and 4.6. The locations of four pulses are also randomized based on a Gaussian distribution. In the remains of this subsection, the simulated noisy EEG waveforms are used to test performance of the three models described above and results are shown in Fig. 6. It is not hard to see that each method can remove OAs from EEG recordings well. Especially, DWT-Kalman filtering protects EEG in non-OA areas whereas the other two methods do not.

To quantitatively compare OAs removal effects of the three models, we evaluate de-noising performance in time domain and frequency domain respectively. In time domain, we take the index MSE [34] between true EEG and corrected EEG to assess performance of different methods. The MSE is defined as (9). N , equaling 5120, is the total samples of each waveform; $x(k)$ represents true EEG signal and $\hat{x}_k$ denotes corrected EEG signal. The result derived from all 50 pairs of waveforms averaged is shown in table I. Obviously, MSE of DWT-Kalman is 0.0017, significantly less than that of WPT-ICA and DWT-ANC, which is 0.0468 and 0.0052 respectively, so DWT-Kalman filtering is superior to the other two methods.

TABLE II. DIFFERENT METHODS' MSE VALUE

Value	WPT-ICA	DWT-ANC	DWT-Kalman
MSE	0.0468	0.0052	0.0017

TABLE I. DIFFERENT METHODS' MAE VALUE AT DIFFERENT FREQUENCY BAND

Value	Band	WPT-ICA	DWT-ANC	DWT-Kalman
MAE	δ -band	0.0266	0.0265	0.0094
	θ -band	0.0269	0.0153	0.0082
	α -band	0.0223	0.0036	0.0029
	β -band	0.0113	0.00061	0.0004
mean value		0.0218	0.0115	0.0052

$$MSE = \frac{1}{N} \sum_{k=1}^N (x(k) - \hat{x}(k))^2 \quad (9)$$

$$MAE = \frac{1}{j-i} \sum_{n=i}^j |P_y(n) - P_{\bar{y}}(n)| \quad (10)$$

The MAE [34] for calculating frequency difference between true data and corrected data is used to evaluate performance in frequency-domain. Here, it is defined for each of the four bands: δ -band ($f < 4\text{Hz}$), θ -band ($4\text{Hz} \leq f < 8\text{Hz}$), α -band ($8\text{Hz} \leq f < 13\text{Hz}$) and β -band ($13\text{Hz} \leq f < 30\text{Hz}$). MAE is defined as (10). Where $P_{\bar{y}}$ and P_y , which are calculated via Burg's method, are power spectral density of corrected EEG and true EEG respectively. i and j indicate frequency range of a particular band. The MAE value of each approach in different frequency bands is given in table II, similarly, each value is the result of 50 pairs of waveforms averaged. From the definition, we can see that smaller MAE value means better de-noising performance. As the table shows, index belonging to DWT-Kalman is smaller than the corresponding values obtained from other two methods, which suggests the new method has good performance for minimizing amplitude of OAs and recovering true EEG.

In order to test usability of the method in practice, two groups of data are collected. One set is using braincap, which is produced by BRAIN PRODUCTS; and the other set is using a three channel EEG collector, which is developed by our team. Then, both data set are used to evaluate performance of our method.

B. Test with Data Collected by Three-channel EEG Collector

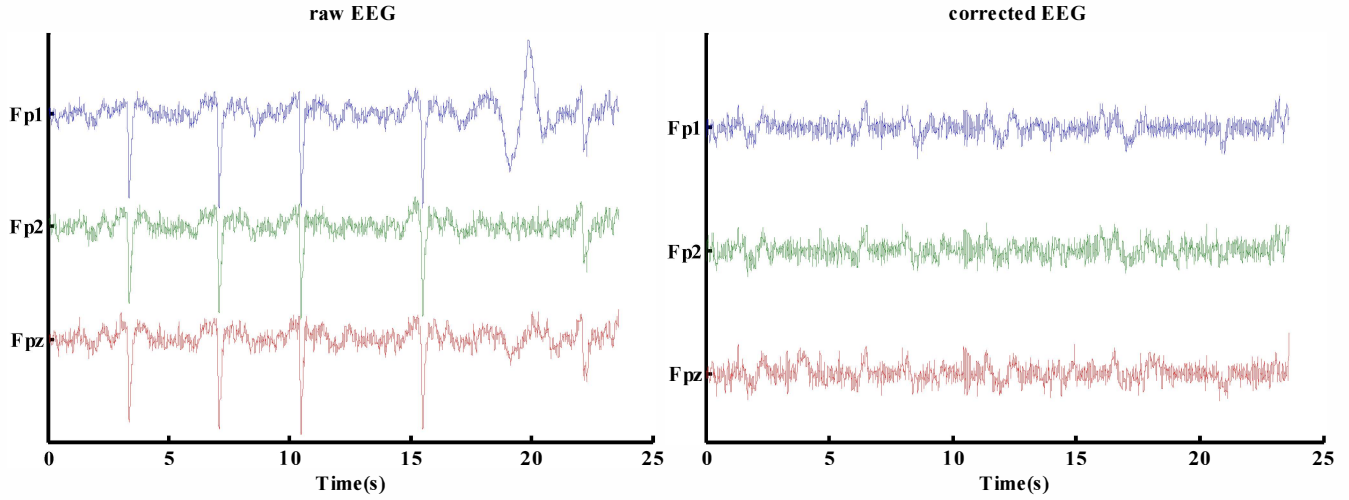


Fig. 8. Left: raw EEG collected by three-channel EEG collector. Right: de-noised results of the left.

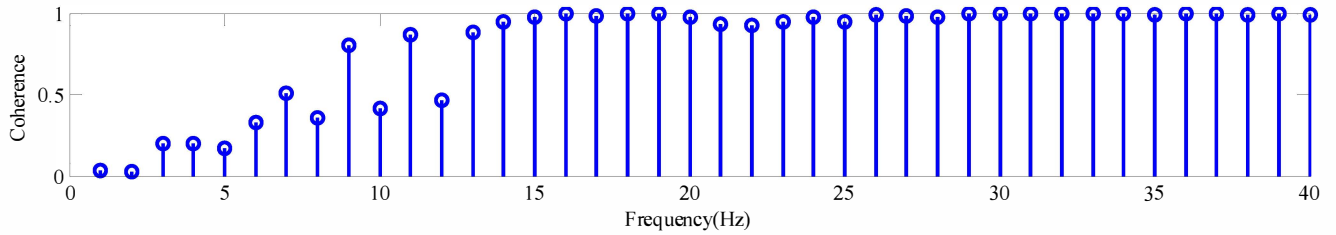


Fig. 9. Magnitude squared coherence estimate of raw EEG and corrected EEG at Fp1.

This group data is recorded by three-channel EEG collector developed by our team, which is part of work of our current project. The three-channel EEG collector is displayed in Fig. 7. The EEG signals are recorded from three electrodes Fp1, Fp2 and Fpz, according to a standard international 10-20 system referenced to earlobes. EEG data is sampled at 250Hz. In order to avoid the influence of base line drift and power line interference, we only select signals in the range of 0.5Hz to 40Hz. Now, we apply our method to the data, and result is shown in Fig. 8. Not hard to see, the method can effectively eliminate OAs from EEG.

To further analyze over frequency domain, magnitude squared coherence estimate between corrected EEG and raw EEG at Fp1 is computed. Magnitude squared coherence estimate between 0 and 1 indicates the quality of two signals x and y , corresponding at various frequencies [6]. It is a function of power spectral of two signals (x is raw EEG while y is de-noised EEG) and is mathematically computed as (11)

$$C_{xy}(f) = \frac{P_{xy}(f)^2}{P_{xx}(f)P_{yy}(f)}, \quad (11)$$

where $P_{xy}(f)$ represents cross power spectral density of raw EEG and corrected EEG, $P_{xx}(f)$ indicates power spectral density of raw EEG and $P_{yy}(f)$ is that of corrected EEG, all of them are computed using Welch's averaged modified periodogram method. As shown in Fig. 9, the correlation of

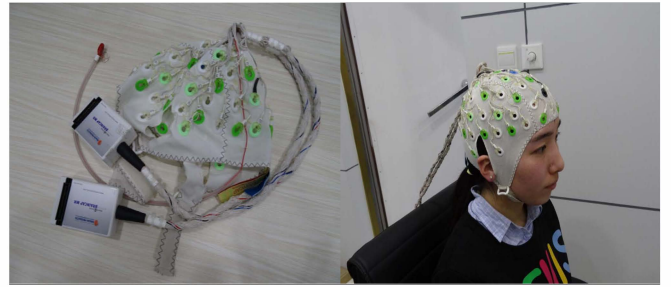


Fig. 10. 64-channel braincap produced by BRAIN PRODUCTS.

low-frequency region below 13Hz in particular is weak whereas the correlation of high-frequency region above 13Hz is strong. Hence, this approach just processes EEG in low-frequency band but has no effect on EEG in high-frequency band.

C. Test with Data Collected by Braincap

The 64-channel braincap produced by BRAIN PRODUCTS, shown in Fig. 10, was used to continuously record EEG signals from subjects in our projects with sample rate of 1000Hz. Signal acquired at forehead is always contaminated most severely, so the proposed method is applied only to channels Fp1, Fp2, and Fpz. For convenience of processing, we down-sample the raw EEG, from 1000Hz to 250Hz. In addition, we also filter the down-sampled EEG in the range of 0.5Hz to 40Hz. Then, we apply the method to the processed EEG data. The result is displayed in Fig. 11. We can see that this method remove OAs from EEG effectively.

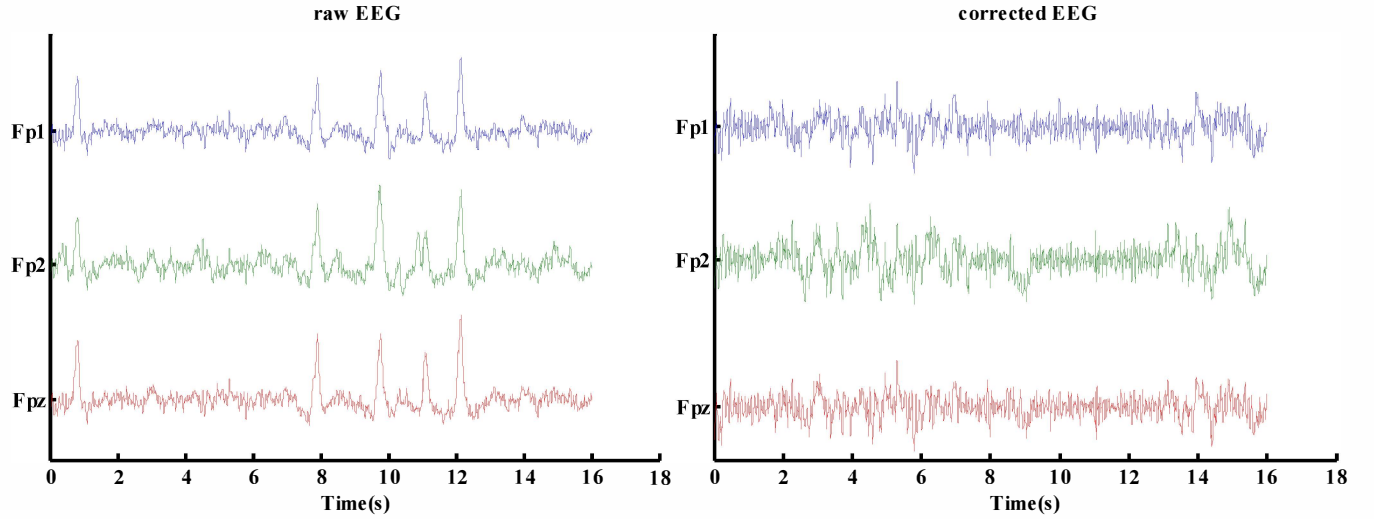


Fig. 11. Left: raw EEG collected by braincap. Right: de-noised results of the left.

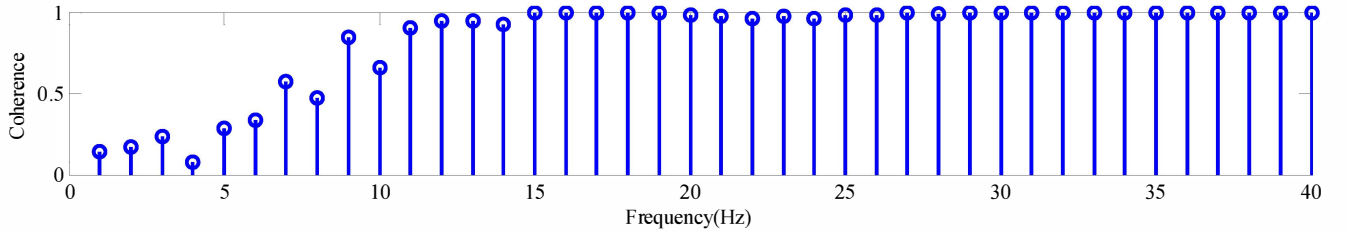


Fig. 12. Magnitude squared coherence estimate of raw EEG and corrected EEG at Fp1.

Furthermore, we calculate the magnitude squared coherence estimate of corrected EEG and raw EEG at Fp1, the result is presented in Fig. 12. We get similar conclusion as previous: the correlation of low-frequency region below 11Hz is weak whereas the correlation of the high-frequency region higher than 11Hz is strong.

IV. DISCUSSION

In this paper, we have put forward a method combined with DWT and Kalman filtering. In this hybrid model, OA zones are firstly detected and extracted separately in order to preserve EEG in non-OA area. Afterwards, wavelet is applied to decompose OA segments, and the OA segments are roughly reconstructed just using the coarsest coefficients. Then, we apply Kalman filtering to the approximated OA fragments to acquire pure OAs, which is finally subtracted from raw EEG recordings. Hence, pure EEG signals are obtained. Through the experiments with simulated data and real data, we can see that the MSE of our method is 0.0017, significantly lower than that of WPT-ICA and DWT-ANC, which is 0.0468 and 0.0052, respectively. Meanwhile, the MAE of our method is smaller than that of the other two techniques. Besides, we apply our method to real data collected by three-channel EEG collector and braincap. Both achieve satisfying results. It shows that our new method can eliminate OAs from EEG recordings efficiently.

Comparing with other methods, our new model can efficiently remove OAs from contaminated EEG recordings. Since the application of this method avoids EOG channel for

collecting synchronous ocular reference signal, it facilitates the use of portable equipment. Although there are still some aspects that can be improved in the proposed model, such as operating efficiency, the result of OAs reduction is satisfying for application in practical projects.

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